Luna Wu Cavin Dales

Nottingham, NH

Class of 2027 | Virtual Learning Academy Charter School, Exeter, NH

EDUCATION

Virtual Learning Academy Charter School, Exeter, NH

High School Diploma, Expected 2027

Cumulative Unweighted GPA: 4.00 / 4.00 | Designation: NH Scholar STEM Pathway

SAT: 1530 (780 Math, 750 Reading and Writing)

Class Rank: top 10% (school only provides decile)

Intended Major: Aerospace Engineering | Additional Interest: Computer Science

Secondary Interest: Human Factors Design | Adaptive Technology

Advanced Coursework

- AP courses: Physics 1 (Exam Grade – 4)
- AP Computer Science Principles (Exam Grade – 4)
- AP Computer Science A (in progress)
- AP Calculus BC (in progress)
- AP English Language and Composition (in progress)
- AP Macroeconomics (in progress)
- Honors Precalculus
- Honors Algebra 1-2
- Honors Geometry
- Honors Biology
- Honors Chemistry
- Honors Physical Science
- Honors English 1-3 Honors
- Honors U.S. History
- Honors Mandarin Chinese 2-3
- Honors Creative Coding JavaScript 1

ENGINEERING AND TECHNOLOGY PROJECTS

Space Why? Rocket Project | Independent Project

- Developed a flight computer (Flying Blind) through component selection, schematic and printed circuit board (PCB) design, assembly, testing.
- Wrote flight computer software to control onboard barometric, gyroscopic, and acceleration sensors, and MOSFET, servo control, and data storage components as well as calculate angular velocity, multiple axes positioning, and altitude.
- Implemented algorithms for Kalman Filter Altitude Estimation and Proportional-Integral-Derivative (PID) stabilization.
- Developed and tested thrust vector control (TVC) hardware, software and guidance functionality.
- Modeled, simulated and analyzed rocket performance using MATLAB, Simulink, and Open Rocket
- Applied soldering, Schematic/PCB Design, PCB assembly, and 3D printing skills to project hardware.
- Project details and repository: https://github.com/luna-dales

Where is it? Accessible Shopping App | Independent Project | Summer 2026-Present

- Developing an iPhone app to help people with low vision locate products and navigate unfamiliar stores; developed preliminary app designs and layouts.
- Researched product-location information, store maps, public API availability, and planograms across Publix, Hannaford, Shaw's, Lowe's, Home Depot, and Walmart.
- Created crowd-sourced AI based product location technology using image recognition, GPS data
- Revised the design to accommodate differences in the location information available from each store chain.
- Sought input from NOAH (National Organization for Albinism and Hypopigmentation) peers and discussed opportunities to help or test the app; incorporated in-store navigation into the design following feedback.
- Exploring camera assistance for identifying products on shelves and an AI search assistant; these features remain in the planning stage.

VOLUNTEER EXPERIENCE

Crocheter and Volunteer | Warm Up America Foundation

- Crocheted 50 blanket squares, 10 hats, and 10 pairs of gloves for donation to people in need, as well as many amigurumi animals.

Volunteer | Suncoast Human Society of Englewood Florida

- Served as cat caretaker and attendant
- Helped construct new shelter and cat rescue center
- Built furniture and equipment for cat center

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Martial Arts | Student and Peer Instructor | High School

- Train in two martial arts disciplines at different dojos, progressing through several belts and practicing with both children and adults.
- Help teach younger students by explaining and demonstrating techniques and recognizing when a different approach is needed.
- Develop situational awareness, reflexes, and adaptive techniques to compensate for limited depth perception.

National Organization for Albinism and Hypopigmentation | NOAH

- Led an ongoing teen book club.
- Crocheted an amigurumi giveaway prize for a NOAH conference fundraiser.
- Participate in annual national conferences.
- Co-creator and instructor of a low-vision sailing workshop.

ADDITIONAL EXPERIENCES AND PROGRAMS

Space Camp for Interested Visually Impaired Students | SCIVIS

- Served as Commander, Mission Specialist, and Flight Director in simulated International Space Station missions.
- Participated in adapted astronaut training, including the Multi-Axis Trainer, 5 Degrees of Freedom Chair, and Underwater Astronaut Trainer.
- Developed an interest in designing spacecraft and interfaces that accommodate a broader range of human abilities.

Summer Academic and Leadership Programs

- Coding Academy: Intro to Python | Harvard University

- Business Academy: Entrepreneurship | Harvard University
- National Student Leadership Conference: Aerospace Program | Duke University

AWARDS AND RECOGNITION

- College Board National Recognition Program - Rural and Small-Town Recognition Award (2026)
- NH Scholar Stem Pathway designation
- National Association of Rocketry – L1 High Powered Rocketry Certificate
- NAUI Open Water SCUBA Diver Certificate

SKILLS AND INTERESTS

- Aerospace engineering, human factors, accessible spaceflight
- Accessibility research, adaptive mobile application layout design
- Peer instruction, collaboration, and problem-solving
- Martial arts